Manas Dutt

AI / SOFTWARE ENGINEER | GENERATIVE AI | BACKEND

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SUMMARY

Computer Science undergraduate at VIT Chennai with hands-on experience building **AI-driven applications** across generative AI, multimodal LLMs, RAG/vector search, computer vision, and backend systems. Experienced with **LangChain, Gemini, Llama 2, Pinecone, PyTorch, OpenCV, Streamlit** and AI-assisted development workflows.

EXPERIENCE

Software Engineering Intern **May 2025 – Jul 2025**

Hetu Labs

- Engineered backend utilities reducing API response latency by **25%**; contributed to scalable services and CI/CD workflows.
- Refactored utilities into modular components and improved documentation for maintainability and reuse.

Summer Research Intern (SRIP) **May 2026 – Jul 2026**

VIT Chennai

- Developed a **CNN-based Deepfake Detection System** using **Python, PyTorch, and OpenCV**, covering preprocessing, training, and evaluation.
- Conducted faculty-mentored computer-vision research and documented experimental findings.

SELECTED AI PROJECTS

Invoice Intelligence Extractor [GitHub] *Python, Gemini 2.5 Flash, Streamlit*

- Built a multimodal **LLM-powered invoice analyzer** using Gemini 2.5 Flash for image-based Q&A and information extraction.
- Implemented **structured JSON extraction** for invoice number, date, vendor, total, and tax fields.

AI-Driven Learning Style Prediction [GitHub] *Python, KAN, TabNet, SS-VAE, CatBoost, Gemini*

- Built an **AI-driven learning style prediction system** using KAN, TabNet, SS-VAE, and CatBoost to analyze **18 behavioral features**.
- Integrated **Google Gemini** to generate personalized AI-powered study plans from predicted learning profiles.

Semantic Document Search & PDF Q&A [GitHub] *LangChain, Pinecone, Gemini*

- Built a **RAG pipeline** using document chunking, sentence-transformer embeddings, Pinecone vector retrieval, and LLM generation.
- Enabled natural-language Q&A over unstructured documents through semantic search and retrieval.

Shadow Detection & Removal [GitHub] *PyTorch, OpenCV*

- Improved DESDNet by **18% IoU** and optimized inference across **10,000+ satellite images**.

SOFTWARE

MSME Invest Connect [GitHub] *React, Node.js, Express.js, MongoDB*

- Built a full-stack investment platform with JWT-authenticated REST APIs and optimized MongoDB queries for backend performance.

TECHNICAL SKILLS

GenAI: LLM Applications, Prompt Engineering, LangChain, Llama 2, Gemini, RAG, Embeddings, Vector Search, Multimodal AI

ML: PyTorch, TensorFlow, Scikit-learn, CatBoost, SHAP, OpenCV, CNNs, SMOTE **Languages:** Python, Java, C++, JavaScript, SQL

Backend: Node.js, Express.js, Flask, React.js, REST APIs **Data/Tools:** Pinecone, ChromaDB, MongoDB, MySQL, Firebase, Docker, Linux, Azure, Git

AI-Assisted Development: Rapid prototyping, code generation/refactoring, LLM-assisted debugging and documentation

EDUCATION

Vellore Institute of Technology, Chennai **Expected May 2026**

B.Tech in Computer Science and Engineering *CGPA: 8.74/10*

Class XII (CBSE): 97% **Class X (CBSE): 93.4%**